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HUMN-391-02
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18 May 2026

Project #2: Aerospace Tooling Community Discourse

Engineering is a broad field with a lot of different corners to it, and most people outside of it would probably struggle to describe what happens in all of them. Aerospace tooling and manufacturing engineering is one of those corners that does not get talked about much, but it is genuinely fascinating once you start looking into it. These are the people who design and build the tools, fixtures, and jigs that make it possible to manufacture aircraft and spacecraft components with the kind of precision those applications demand. Without them, the composites and assemblies that go into modern aircraft could not be produced consistently or safely. It is critical work that sits just out of view behind the more visible parts of aerospace, and we think that is part of why it appealed to us as a community to study.

We had the opportunity to sit down with an experienced aerospace tooling engineer who has spent his career leading design and supporting complex tooling for aerospace manufacturing programs. Going into the interview, we had a general sense of what aerospace manufacturing involved from our coursework. But, coming out of it, we had a much deeper appreciation for how complex and interconnected this community really is. A lot of what he described surprised us, not because it was strange, but because it was more normal than we expected. The technical standards are rigorous, but what actually holds this community together is something more like culture and reputation than certification and paperwork.

A discourse community is a group of people who share common goals, values, writing standards, a specialized vocabulary, and specific genres of communication. Aerospace tooling fits that description in ways that are both formal and cultural. Formally, the community is held together by certifications and quality standards that define who can even participate in the work. The cultural side is harder to define, but it comes down to reputation and trust, the kind that only accumulates over years of showing good judgment and being honest with people when things go wrong.

One of the most interesting things we learned from our interview is that getting a degree is really just the starting point in this community. Most engineers have mechanical engineering backgrounds, but our interviewee was clear that it is not the only path in. Some of the best designers he has worked with came up through tool and die or machining, learning the design side entirely on the job. Others started as welders or machinists and developed into designers over time because they understood at a base level what was actually buildable. That practical

knowledge carries real weight in a way we found genuinely interesting compared to how credentials tend to work elsewhere.

The certifications that matter most are held by companies, not individuals. AS9100 Rev D and ITAR registration are what our interviewee described as the cost of entry. Without them, a company cannot do aerospace tooling work at all. Individual certifications like a PE license, which matter so much in fields like civil engineering, are actually rare in aerospace. What matters at the individual level is harder to earn and harder to fake. It is whether you have been through a full program cycle from start to finish, quoting, design, fabrication, inspection, buyoff, and supporting the tool once it is in service. An engineer who has done that entire cycle has standing in the community that someone who has only seen pieces simply does not yet have.

Our interviewee put it pretty directly: "A young engineer with a degree but no program experience will be respected, but they won't be trusted yet." Trust comes from demonstrated judgment over time. Knowing when a tight tolerance is actually necessary and when it does not matter. Knowing when to push back on a customer requirement that does not make technical sense. And being honest when something has gone wrong rather than trying to manage how it looks. Getting caught lying in a design review, he told us, is one of the fastest ways to destroy your credibility in a community that values directness above almost everything else. "If you can't explain your rationale in plain language to a machinist, a customer, and a program manager, you'll struggle regardless of your technical credentials." That stuck with us, because it gets at something we have already started noticing in our own co-ops and coursework. Technical knowledge is necessary but it is not the whole picture.

Plain language in this context does not mean eliminating technical vocabulary entirely. It means using the right level of detail and vocabulary for the audience in front of you. A machinist does not need a tolerance stack-up derivation; they need to know the critical dimension and why it matters to the fit. A program manager does not need the weld sequence; they need to know whether the schedule is at risk. Plain language is about clarity of intent, not simplicity of content. Our interviewee made clear that fluency in technical terms is still expected, but using jargon as a shield rather than a tool is what causes problems. There is also a distinction worth drawing between withholding information and communicating strategically. Not every detail belongs in every conversation. A shop-floor discussion does not require a full contractual risk breakdown, and leading a design review does not mean reading every analysis aloud. What crosses the line into credibility-destroying territory is misrepresenting facts, claiming certainty you do not have, or staying silent about a known problem in a way that causes real harm downstream. That is different from choosing which level of detail is appropriate for your audience.

Inside the organization, the most frequent communication still happens face-to-face. Our interviewee described a designer walking out to the weld shop to talk through a fabrication sequence with the shop lead, where much of the real engineering actually gets done. It is fast, iterative, and both people can point at actual hardware or a printed drawing while they talk. The conversations in those settings are short and informal because everyone shares the same

technical context. He gave us a simple example, describing an internal conversation as something like, "that weld joint is going to pull like hell when it cools." No formality, just two people solving a problem in real time.

With a customer's engineering team, the same concern sounds completely different. "We've identified a risk of weld distortion at this joint. Our manufacturing plan includes distortion mitigation through compensated weld sequencing and post-weld stress relief, and we'll verify final flatness with laser tracker inspection per the drawing requirements." The technical content is the same, but the framing, the vocabulary, and the level of formality are completely different. We thought that contrast was one of the clearest illustrations of what adapting language to different audiences actually looks like in practice, and it is the kind of thing that is easy to describe in theory but much harder to actually do well.

Email handles anything that needs a written record, especially anything involving customers. If something goes wrong later, you can point to the record, and in an industry where contracts and regulations aerospace hold significant weight, that paper trail is more of a requirement than just professional courtesy. Microsoft Teams or similar platforms handle quick internal communication that needs a faster answer than an email. PowerPoint is the expected format for any structured presentation to customers or internal leadership. One thing that stood out to us was how much mode of communication shifts based on the sensitivity of the information. ITAR-controlled technical data does not go through a Teams message or a casual email chain. It moves through controlled channels with verified recipients, and quality issues with customers get handled through formal Non-Conformance Reports with defined distribution lists, not phone calls.

One of the things we asked about directly was what information flow actually looks like across a full program, from an initial customer requirement all the way through to a delivered tool. The answer revealed just how many different groups are involved and how the genre and mode of communication shifts at every handoff.

It starts with the customer's engineering team issuing a Statement of Work along with a part model or drawing. That package goes to program management and estimating, who develop a quote with input from the design lead and manufacturing engineering. Once the program is awarded, the design team develops the tool concept and runs it through the customer, incorporating feedback and closing out action items before the design is finalized. The approved design produces a drawing package that goes to manufacturing engineering for process planning. From there, it goes through the shop floor in stages: weld fabrication first, then rough machining, stress relief, and finish machining to final dimensions. Quality and inspection verify dimensions throughout and produce the FAI (First Article Inspection) package at the end. The completed tool ships to the customer for physical buyoff, often with the contractor's engineers present.

Our interviewee pointed out that at each of these handoffs, the genre and mode of communication shift. Customer-to-contractor is formal statements of work and specifications. Internal design work is CAD (Computer-Aided Design) models and informal technical

discussion. Design-to-shop includes drawings and manufacturing plans. Shop-to-quality is inspection data. Contractor-to-customer at buyoff is the FAI package and a physical demonstration. Understanding that map and knowing which genre belongs at which handoff is something that takes real time to develop, and it is one of the things that distinguishes someone who has been through full program cycles from someone who has not. The map we developed for this project tries to capture that full flow visually: from the customer engineering team issuing a SOW at the top, through program management and the design lead, down through the design team producing CAD models, 2D drawings, and PDR/CDR review cycles, then into manufacturing engineering, the shop floor, quality and inspection, and finally customer buyoff. The key genres and communication modes at each stage are what make the handoffs work, and fluency in that system is what separates engineers who have seen full program cycles from those who have not.

The genres of writing this community produces form a system, each one serving a specific audience at a specific point in a program's life. The drawing package is the foundation. Two-dimensional production drawings per ASME Y14.5, generated from three-dimensional CAD models, define exactly what gets built and to what tolerances. Precision is non-negotiable. A single ambiguous dimension or an unclear datum reference can create real problems across an entire fabrication sequence.

FAIs, governed by AS9102, document that every characteristic on the drawing was measured and met spec on the first completed tool. They are the formal proof that you built what you said you would build. Non-Conformance Reports, or NCRs, document anything that did not meet specification on a given build. When a dimension is out of tolerance, a material is wrong, or a process was not followed correctly, an NCR is opened. The NCR records what the nonconformance was, who found it, and what disposition was chosen: use-as-is if the deviation has no functional impact, rework to bring it back into spec, or scrap and rebuild. The NCR system is not just paperwork. It creates accountability and a traceable record that quality engineers, program managers, and customers can all reference. Without it, the same nonconformance can reappear on the next build because nobody documented what happened or why the disposition was acceptable. Engineering Change Orders and Notices capture any design change after a baseline was established and are how the organization keeps everyone building to the same version of a design.

Failure analyses are more narrative than the other documents in this system. When a tool fails in service or a quality escape occurs, the failure analysis documents what happened, why, and what is being done to prevent recurrence. Our interviewee described these as needing to be clear enough that people without engineering backgrounds can understand the findings and evaluate the corrective action. That is a different kind of writing challenge than producing a drawing or an FAI. Manufacturing plans translate the drawing package into step-by-step build sequences written for the welders, machinists, and assemblers who will do the actual work.

Design review presentations are where a lot of these loose ends come together and where an engineer's credibility is most visibly on the line. Our interviewee told us that the worst thing you can do in a design review is make something up when you do not know the answer. The right move is to say you do not know and commit to following up. Customers, he said, respect honesty far more than hand-waving. That is a piece of advice that translates pretty directly to a lot of other professional situations we will find ourselves in.

Aerospace tooling has its own vocabulary, and fluency in it is both a practical necessity and a symbol of belonging. Acronyms like FAI, NCR, ECO, PDR, and CDR are standard across aerospace manufacturing. Aircraft-specific terms like stringers, bulkheads, ribs, and spars are all used solely within the aerospace industry.

Within the tooling community specifically, the vocabulary reflects the materials and processes that define the work. Invar 36 is the nickel-iron alloy used for tool faces because its coefficient of thermal expansion (CTE) nearly matches carbon fiber composite, meaning the tool and the part grow together during the cure cycle rather than fighting each other thermally. CTE comes up constantly because managing thermal behavior is one of the defining technical challenges of the work. Laser trackers and CMM (Coordinate Measuring Machines) refer to the precision measurement equipment used to verify dimensions on large tools. Datum structure, buyoff, stress relief, and weld distortion all carry precise meanings that take real time to absorb and use correctly.

On top of the industry-wide vocabulary, every organization has its own internal layer of program names, customer codes, and project designations that mean nothing outside the building. A new engineer has to learn both layers simultaneously, which is part of why fitting into this community takes as long as it does. We have experienced a version of this in our own co-ops, where the first few weeks involve as much vocabulary acquisition as anything else. In aerospace tooling, that curve is very steep.

Something we did not expect to spend as much time on in the interview was the design and manufacturing philosophy that shapes how work gets done in this community, and it turned out to be one of the most interesting parts of the conversation.

Aerospace tooling is almost never high volume. A company might build one tool, or a set of five fixtures, for a specific program. That means every unit carries the full burden of setup costs, programming costs, and material costs. You design for the process you have, minimize setups because each one costs time and introduces alignment risk, and think carefully about weld sequences that minimize distortion so you do not spend expensive machining time chasing flatness afterward.

Material utilization also matters significantly. Invar 36 runs roughly five to eight times the cost of carbon steel, is difficult to machine, gummy, prone to work hardening, and hard on cutting tools. The design philosophy that comes out of this is to limit Invar content to the surfaces where the CTE match actually matters, use steel for the structural elements underneath,

plan rough machining allowances carefully, and stress relieve after welding and between rough and finish machining passes. That last step adds schedule time but is non-negotiable for dimensional stability.

What struck us most about this part of the conversation was how much upfront investment goes into getting things right digitally before anything gets cut. Tolerance stack-ups, thermal simulation, interference checks, sometimes finite element analysis, all of that happens before material is ordered. In high-volume manufacturing, you can discover and correct problems over successive production runs. In low-volume aerospace tooling, you might build something once and never build it again. The first one has to come out right.

Translating technical information for people who do not share the community's vocabulary is one of the most important and least formally taught skills in this field. With customers, communication flows through defined channels rather than directly between a designer and the customer's engineer, because anything put in writing to a customer can become a contractual commitment. The program manager is usually in the loop, partly for relationship management and partly because the documentation created in those exchanges becomes part of the program record.

Communicating with the shop floor is its own kind of translation. Manufacturing plans are written for welders, machinists, and assemblers, not engineers with four-year degrees, and being able to do that clearly is a genuine professional competency. Working with outside subcontractors is tightly controlled by legal structure. Subcontractors operate under flowdown NDAs and, if the work is ITAR-controlled, must themselves be ITAR-registered. Information is shared strictly on a need-to-know basis.

ITAR shapes external communication in ways that have no real parallel in most other engineering fields. The categories of restricted information include detailed design data for defense-related tooling, manufacturing process parameters, material specifications tied to specific programs, and sometimes even the existence of certain programs. Collaboration with outside organizations is always filtered through an awareness of what can and cannot leave the building, and that is not just a cultural norm. It is a legal obligation with real consequences.

If there is one thing that came through most clearly in our interview, it is that this community's values are enforced through culture and reputation just as much as through formal systems. When something does not meet spec, the NCR process requires formal documentation and a defined path to disposition. Nobody has unilateral authority to ship a non-conforming tool without going through that process.

Schedule pressure is real and constant. Programs have contractual delivery dates, while cost overruns on fixed-price contracts hurt. The interviewee acknowledged that pushing for creative solutions is appropriate and sometimes there is a rework path nobody has considered. But the quality system exists precisely to make sure that pressure does not override the engineering call.

What really reinforces all of this is how small the aerospace tooling world is. Everyone eventually knows everyone else, and your reputation travels with you. The reality that one shortcut can undo years of trust creates strong incentives to do things the right way, not just because people are inherently honest, but because the cost of compromising integrity in a community this size is genuinely high.

Going into this project, we knew aerospace was a demanding field. What we did not fully appreciate was how much of what makes the aerospace tooling community work has nothing to do with technical knowledge, but everything to do with how people earn trust and communicate across very different audiences. The formal systems, the quality standards, the documentation requirements, all of that exists for a reason. But underneath it is a professional culture that is consistent in what it values: be honest, know your stuff, understand what happens when your design hits the shop floor, and never ship something you know is wrong. For us as engineering students heading into manufacturing-adjacent fields, that is probably the most transferable thing we took out of this. The technical skills are learnable. The judgment and the integrity are what actually determine how far you go.